

Project Blueprint: Brew Corner Café Full-Stack Web Application

1. Executive Summary & Client Context

- **Business Name:** Brew Corner Café
- **Location:** Jaro, Iloilo City, Philippines
- **Business Contact:** 0917-123-4567 | brewcorner@example.com
- **Operating Hours:** 8:00 AM – 8:00 PM (Daily)
- **Core Concept:** A small, modern neighborhood café serving coffee, snacks, and pastries. The goal is to build a dual-system platform consisting of a responsive customer-facing ordering website and a non-technical staff-friendly administration dashboard.

2. Technical Stack & Tools

- **Frontend Framework:** Next.js (React) with App Router
- **Styling:** Tailwind CSS (Theme: Warm brown, cream, modern minimalist aesthetics)
- **Backend & Database:** Supabase (PostgreSQL)
- **Realtime Communication:** Supabase Realtime (for live admin order queues)
- **Authentication:** Supabase Auth (Role-based: Admin vs. Staff)
- **Storage:** Supabase Storage Buckets (for menu item images)
- **Hosting & Deployment:** Vercel

3. System Architecture & Database Schema

The architecture utilizes a serverless decoupled approach. Next.js communicates directly with Supabase via client-side/server-side SDKs, removing the need for a separate API server.

4. Development Implementation Plan

To maximize development velocity and ensure robust data flow, development will be executed in three structural phases.

Phase 1: Database Setup & Infrastructure (Current Step)

- Initialize Supabase project environment.
- Execute database schemas and set up Row Level Security (RLS) policies.
- Create a Supabase Storage Bucket (`menu-images`) for image hosting.
- Seed initial database categories (`Coffee`, `Non-Coffee`, `Pastries`, `Desserts`) and staff accounts via Supabase Auth.

Phase 2: Back-Office Admin Dashboard

- Build custom login portal tailored for `admin` (Anna - Owner) and `staff1` (Mark - Staff).
- Create **Menu Management Panel**: CRUD functionality for menu items, toggling stock availability (`is_available`), and automated image uploads.
- Create **Real-time Order Queue Dashboard**: A reactive board leveraging Supabase WebSockets to stream incoming orders instantly, rendering distinct interactive states (`Pending > Preparing > Ready > Completed / Cancelled`) accompanied by audible alerts.

Phase 3: Customer-Facing Front Window & Checkout Flow

- Design responsive frontend using Tailwind CSS matching the required palette (`#FDFBF7` cream bases, `#78350F` warm amber/brown accents).
- Implement landing UI details: About text, operational metadata, contact parameters, and an 8–10 asset image gallery grid.
- Develop reactive shopping cart engine via localized context state (quantity tracking, modifier previews).
- Implement secure checkout form handling delivery configurations (injecting dynamic ₱50 fee calculations for Iloilo City parameters) and confirmation feedback pages.

5. Global Scope Constraints & Business Rules

- **Delivery Fee**: Fixed at ₱50 for addresses located inside Jaro / Iloilo City.
- **Fulfillment Window**: Preparation baseline targets 20–30 minutes per queue entry.
- **Payment Handlers**: Constrained purely to Cash on Delivery/Pickup or manual GCash references.
- **UX Requirement**: Admin viewports must prioritize hyper-legible, high-contrast layouts tailored explicitly to accommodate staff members unfamiliar with complex tech tools.